Wind Analysis Design Calculations

Project Name:	**Supply and Installation of Manual Sliding Doors**				**NTPL Project Number:**	NTPL-72-PM
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WIND ANALYSIS REPORT

DOC No	Company	Zone	Line	Doc type	Discipline	Sequence	Rev	Version	Status

NIHVA TECHNOLOGIES PVT. LTD.	LOCATION - POSCO MANGAON Wind Analysis for Sliding Door	Project No: NTPL-72-PM Mfg. Year: June 2022

Max Door Size 7300 W x 5000 H mm: Manual Sliding Door Four Leaf - RSS

Dead Load

Door size = 7.3 m x 5.0 m = 36.50 sqm

Self-weight of door = 50 kg/sqm x 36.50 = 1825 kg = 1825/100 = 18.25 KN (stability force)

Wind Load

Wind loads on structures shall generally be as per IS: 875-1987-Part 3 based onfollowing parameters: -

Description	Applicable	Remarks
Basic wind speed (Vb)	39 m/s	As per IS-875-1987-Part 3
Terrain Category	2	Assumed Obstructions up to height of 1.5m to 10m.
Risk coefficient (k1)	1.0	As per Table 1 of IS-875-1987-Part 3
Terrain height/structure size factor(k2)	1.05	Shall be suitable taken as per Table 2 of IS-875-1987-Part 3 for relevant class of structure.
Class	A/B/C	
Topography factor(k3)	1.0	Assumed as per Cl.5.3.3 of IS-875-Part3

For computation of wind forces the mean probable design life of all structures shall be taken as 50 years.
Temporary structures shall be designed for a mean probable design life of 25 years and boundary walls as per IS: 875-Part 3
Design wind speed = Vz = k1 xk2x k3 x Vb
Design wind pressure = Pz = 0.6xVz2

If Building is closed structure, external pressure coefficient Cpe shall be calculated as per Table IS 875-Part3.
Internal pressure coefficient shall be as per clause 6.2.3.2 IS 875-Part3.

Design wind speed = Vz = 1.0 x 1.05 x 1.0 x 39 = 40.95 m/s

Design wind pressure = Pz = 0.6 x 40.95 x 40.95 = 1.006 kN/sqm

Wind load = 1.006 x 36.50 = kN (Sliding Force)

Load Combinations

Listed below are load combinations for analysis & design of Buildings / Structures.
Basic Load Cases

- DL: Dead Load
- WL: Wind Load

Check for Deflection (Serviceability criteria)

Following Load combinations with un-factored loads are to be used for checking deflection for both RCC & Steel Structures & calculation of safe bearing pressure.
Load combinations to check for deflection and calculation of bearing pressure as per IS-875-1987-Part 5 & IS-456-2000

Load Combinations	**DL**	**LL**	**WL**	**EQ**
DL + LL	1.0	1.0	-	-
DL + WL	1.0	-	1.0	
DL + LL + WL	1.0	0.8	0.8	-
DL + EQ	1.0	-	-	1.0
DL + LL + EQ	1.0	0.8	-	0.8

Stabilizing Force = 18 kN

Sliding Force = 38.69 kN

Resultant force = 38.69– 18.25 = 20.44 kN

Connection Design:

SS202 bolts 4 nos of M16 bolt are used for fixing the sliding door at top.

Tension Capacity of bolt = 56.5 kN

Shear capacity of bolt = 31.4 kN

Resultant sliding force = 20.44 kN < 56.50 kN…. Hence connection design is safe in tension.

Stabilizing force = 18.25 kN < 31.40 kN…. Hence connection design is safe in shear.

NIHVA TECHNOLOGIES PVT. LTD.	LOCATION - POSCO MANGAON Wind Analysis for Sliding Door	Project No: NTPL-72-PM Mfg. Year: June 2022

Min Door Size 3400 W x 3500 H mm: Manual Sliding Door - 3 Leaf - RSS

Dead Load

Door size = 3.4 m x 3.5 m = 11.90 sqm

Self-weight of door = 50 kg/sqm x 11.90 = 595 kg = 595/100 = 5.95 kN (stability force)

Wind Load

Wind loads on structures shall generally be as per IS: 875-1987-Part 3 based onfollowing parameters: -

Description	Applicable	Remarks
Basic wind speed (Vb)	39 m/s	As per IS-875-1987-Part 3
Terrain Category	2	Assumed Obstructions up to height of 1.5m to 10m.
Risk coefficient (k1)	1.0	As per Table 1 of IS-875-1987-Part 3
Terrain height/structure size factor(k2)	1.05	Shall be suitable taken as per Table 2 of IS-875-1987-Part 3 for relevant class of structure.
Class	A/B/C	
Topography factor(k3)	1.0	Assumed as per Cl.5.3.3 of IS-875-Part3

For computation of wind forces the mean probable design life of all structures shall be taken as 50 years.
Temporary structures shall be designed for a mean probable design life of 25 years and boundary walls as per IS: 875-Part 3
Design wind speed = Vz = k1 xk2x k3 x Vb
Design wind pressure = Pz = 0.6xVz2

If Building is closed structure, external pressure coefficient Cpe shall be calculated as per Table IS 875-Part3.
Internal pressure coefficient shall be as per clause 6.2.3.2 IS 875-Part3.

Design wind speed = Vz = 1.0 x 1.05 x 1.0 x 39 = 40.95 m/s

Design wind pressure = Pz = 0.6 x 40.95 x 40.95 = 1.006 kN/sqm

Wind load = 1.006 x 11.90 = 11.97 kN (Sliding Force)

Load Combinations

Listed below are load combinations for analysis & design of Buildings / Structures.
Basic Load Cases

- DL: Dead Load
- WL: Wind Load

Check for Deflection (Serviceability criteria)

Following Load combinations with un-factored loads are to be used for checking deflection for both RCC & Steel Structures & calculation of safe bearing pressure.
Load combinations to check for deflection and calculation of bearing pressure as per IS-875-1987-Part 5 & IS-456-2000

Load Combinations	**DL**	**LL**	**WL**	**EQ**
DL + LL	1.0	1.0	-	-
DL + WL	1.0	-	1.0	
DL + LL + WL	1.0	0.8	0.8	-
DL + EQ	1.0	-	-	1.0
DL + LL + EQ	1.0	0.8	-	0.8

Stabilizing Force=5.95 kN

Sliding Force = 11.97 kN

Resultant force = 11.97– 5.95 = 06.02 kN

Connection Design:

SS202 bolts 4 nos of M16 bolt are used for fixing the sliding door at top.

Tension Capacity of bolt = 56.5 kN

Shear capacity of bolt = 31.4 kN

Resultant sliding force = 6.02 kN < 56.50 kN…. Hence connection design is safe in tension.

Stabilizing force = 5.95 kN < 31.40 kN…. Hence connection design is safe in shear.